MODULE 07 · DOMAIN 2 - CLAUDE MODELS, PROMPTING AND CONTEXT ENGINEERING

Choosing the Model

Tiers and trade-offs, a selection method, effort and routing as levers, and re-testing on every change.

DOMAIN 2 of 7 EXAM WEIGHT 13% MODULE 7 of 38

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There is no best model. There is the lightest model that clears your bar.

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The tiers today

TIER, OCTOBER 2026	LATENCY	USD PER M TOKENS, IN / OUT
Haiku 5.5	Fastest	0.10 / 0.50 to 100K prompt
Sonnet 5.5	Fast	2 / 10
Opus 5.5	Moderate	4 / 20
Fable 5.1	Slower	10 / 50

ON THE EXAM

Versions and prices move. The shape of the trade-off does not. Questions use tiers.

Four things every tier trades

- **Capability:** depth of reasoning, reliability on hard or ambiguous cases.
- **Latency:** heavier tiers start and finish more slowly.
- **Cost:** price per token multiplied by your real volume.
- **Rate limits:** separate per model class; the heaviest tier has the lowest.

> ON THE EXAM
>
> Volume turns small price differences into large bills. Always multiply by the real volume.

The selection method

01
Define
Quality bar, latency, volume, budget.

02
Shortlist
Candidates from the requirements.

03
Test
A representative sample, edge cases included.

04
Choose
The lightest that clears the bar.

05
Route
If difficulty varies across requests.

06
Re-test
On every new version.

ON THE EXAM

Requirements come before candidates, and candidates before tests. The exam asks this order.

Where to start when unsure

ANTHROPIC'S API GUIDANCE

Start capable

- Unsure? Start with Opus
- Fable if Opus at higher effort falls short
- Then measure

ANTHROPIC'S USE-CASE GUIDES

Start light for simple volume

- Ticket routing: Haiku at low effort
- Deep expertise or many categories: Sonnet
- Then measure

THE TIEBREAKER

No conflict: start where the workload points, then let your evaluation set decide.

Effort and thinking: dials before you switch

- **Effort:** low, medium, high, xhigh, max. It shapes all tokens, tool calls and thinking.
- **Defaults:** medium on Opus 5.5 and Haiku 5.5; high on Sonnet 5.5 and Fable 5.1.
- **Adaptive thinking** is on by default for the newest models.
- **Raise effort before jumping a tier;** lower it where evals show quality holds.
- **Low effort in long agent prompts** can skip a search or a check.

> ON THE EXAM
>
> Model choice and effort are two dials. Turn the cheaper one first.

One model, or route between tiers?

ONE MODEL

Uniform work

- Similar difficulty throughout
- Simple to build and monitor
- One evaluation to maintain

ROUTE BETWEEN TIERS

Mixed work

- An easy majority, a hard minority
- Light tier for the majority
- Escalate hard or low-confidence cases

THE TIEBREAKER

Mixed difficulty at real volume: route. Uniform difficulty: one model.

Re-test whenever the model changes

- **New versions change behaviour**, even with identical prompts.
- **Run the regression set** on the candidate before promoting it.
- **Defaults move:** a request with no effort set runs lower on Opus 5.5 than on Opus 5.
- **Retirement notice** is at least 60 days; cloud platforms set their own schedules.
- **Migrate early:** effort sweep, cost and latency re-baselined, prompts re-checked.

> ON THE EXAM
>
> A model upgrade is a release. Treat it like one: test, compare, promote, keep a rollback.

Controls the newest models reject

OLD CONTROL	STATUS	USE INSTEAD
`temperature, top_p, top_k`	Rejected from Opus 4.7	Prompting and effort
`Assistant prefill`	Rejected from Opus 4.6	Structured outputs
`Forced tool_choice`	Rejected on Opus 5.5	auto with strict tools

ON THE EXAM

Lowering the temperature was never the fix for grounding or policy errors. On current Opus it is not even possible.

> An online marketplace screens about 2 million new listings a day against 12 policy categories, each with a clear written definition. Each decision must return within 300 milliseconds.
>
> Borderline listings already go to a human review queue.

Which model choice fits best?

A The most capable tier, for maximum accuracy on every listing

B The lightest tier at low effort, with the category definitions, validated on a representative sample

C The Opus tier at maximum effort, so edge cases are handled carefully

D The Sonnet tier with extended reasoning on every listing

◆ PAUSE AND DECIDE

Which model choice fits best?

A The most capable tier, for maximum accuracy on every listing

Slowest and dearest, at two million a day, for clearly defined labels.

B The lightest tier at low effort, with the category definitions, validated on a representative sample

High volume, tight latency, clear labels, humans for edge cases.

C The Opus tier at maximum effort, so edge cases are handled carefully

Edge cases already go to humans. Maximum effort breaks the latency budget.

D The Sonnet tier with extended reasoning on every listing

Reasoning on every call costs latency the task does not need.

THE SIGNAL

Huge volume, tight latency, well-defined labels, humans on the edge: the light tier, validated.

> Quillon asks Corvane to pick a model for drafting prior-authorisation letters. A developer has already started running the same five letters through every tier.
>
> Nobody has agreed what a good letter is, how long clinicians can wait, or how many letters a month to expect.

What should happen first?

A. Finish running the five letters through every tier and pick the one that reads best

B. Choose Opus, because Anthropic recommends starting there when unsure

C. Define the quality bar, acceptable latency, monthly volume and budget

D. Choose the newest model, because it will be supported for longest

◆ PAUSE AND DECIDE

What should happen first?

A Finish running the five letters through every tier and pick the one that reads best

Testing before the bar is defined, on five letters, judged by taste.

B Choose Opus, because Anthropic recommends starting there when unsure

A sensible starting candidate, but still not a requirement.

C Define the quality bar, acceptable latency, monthly volume and budget

Every later step is measured against these.

D Choose the newest model, because it will be supported for longest

Release date is not a selection criterion.

THE SIGNAL

Testing has started before anyone defined success: go back to requirements.

> Tallyhawk routes complex integration faults to the Sonnet tier at its default effort. Most answers are good, but on the hardest faults reasoning is sometimes shallow and misses a second cause.
>
> These conversations can tolerate a few more seconds of latency. The team is weighing its next move.

What should Corvane try first?

A. Move every support conversation to the most capable tier

B. Raise effort on the complex-fault route, then re-run the evaluation set

C. Add "THINK VERY CAREFULLY" in capitals to the system prompt

D. Lower the temperature for complex faults

◆ PAUSE AND DECIDE

What should Corvane try first?

A Move every support conversation to the most capable tier

Changes all routes to fix one, at the highest price.

B Raise effort on the complex-fault route, then re-run the evaluation set

The cheaper dial, aimed at the one route, with measurement.

C Add "THINK VERY CAREFULLY" in capitals to the system prompt

Shouting can over-trigger; effort is the designed control.

D Lower the temperature for complex faults

Varies wording, not reasoning depth, and is rejected on current Opus.

THE SIGNAL

Shallow reasoning on one route, latency to spare: raise effort there first, and measure.

> Mereside's agent assist runs every request on the Opus tier. About 85% of requests are routine single-policy look-ups; the rest are complex cases spanning several policies.
>
> Advisers complain about waiting, and the finance director about the bill. Quality on complex cases must not fall.

Which change fits best?

A Move every request to the lightest tier

B Keep Opus for everything and shorten the system prompt

C Classify each request: routine look-ups to a lighter tier, complex and low-confidence cases to Opus

D Move every request to the Fable tier to make complex cases even better

◆ PAUSE AND DECIDE

Which change fits best?

A Move every request to the lightest tier

Fixes speed and cost, and lets complex-case quality fall.

B Keep Opus for everything and shorten the system prompt

A small saving that leaves the main cause untouched.

C Classify each request: routine look-ups to a lighter tier, complex and low-confidence cases to Opus

The easy majority gets fast and cheap; the hard minority keeps Opus.

D Move every request to the Fable tier to make complex cases even better

Slower and dearer for everyone, the opposite of the complaints.

THE SIGNAL

An easy majority, a hard minority, and a quality floor on the hard part: route between tiers.

> Tallyhawk switched its support agent to a newer model version over a weekend. On Monday, billing answers were phrased differently and two refund explanations were wrong.
>
> The switch was made because the new version was announced as better. No evaluation was run first.

Which process would have prevented this?

A Run the regression set on the new version, compare it with the current one, and promote only with a rollback ready

B Pin the old version permanently, so behaviour never changes

C Move a growing share of traffic to the new version, without an evaluation

D Read the release notes carefully before switching

◆ PAUSE AND DECIDE

Which process would have prevented this?

A Run the regression set on the new version, compare it with the current one, and promote only with a rollback ready

Catches the regressions before customers see them.

B Pin the old version permanently, so behaviour never changes

Models retire. Pinning delays the problem; it does not solve it.

C Move a growing share of traffic to the new version, without an evaluation

Exposes fewer customers, but nothing measures what broke.

D Read the release notes carefully before switching

Notes describe the model in general, not your workload.

THE SIGNAL

Better in general is not better for you. Test the upgrade on your own regression set.

> Some of Quillon's discharge summaries state medication changes that do not appear anywhere in the record. An engineer proposes lowering the temperature, so the model invents less.
>
> The summaries run on the Opus tier.

What is the best response?

A Lower the temperature to zero for all summaries

B Move the summaries to the Fable tier

C Raise the maximum output tokens so the model has room to be careful

D Restrict claims to the record, require each finding to cite its section, and verify against the source

◆ PAUSE AND DECIDE

What is the best response?

A Lower the temperature to zero for all summaries

Wrong layer, and non-default temperature is rejected on current Opus.

B Move the summaries to the Fable tier

A stronger model still has no instruction to stay grounded.

C Raise the maximum output tokens so the model has room to be careful

Length is not the problem.

D Restrict claims to the record, require each finding to cite its section, and verify against the source

A grounding problem needs grounding controls.

THE SIGNAL

Invented facts are a grounding problem. Fix grounding; sampling knobs are the wrong layer.

> Anthropic has announced that the model version behind Mereside's complaints summaries will retire in about ten weeks. Mereside's change board meets monthly.
>
> The current prompts were tuned for this version.

What should Corvane plan?

A. Wait until the final fortnight, then switch to whatever is newest

B. Ask Anthropic to keep the version running for Mereside

C. Start now: evaluate a successor, sweep effort, re-baseline cost and latency, re-check prompts, then switch before retirement

D. Switch production to the newest model today, and fix problems as they appear

◆ PAUSE AND DECIDE

What should Corvane plan?

A Wait until the final fortnight, then switch to whatever is newest

No time to test, and a monthly change board makes it worse.

B Ask Anthropic to keep the version running for Mereside

Retirement dates apply to everyone. Plan for the date.

C Start now: evaluate a successor, sweep effort, re-baseline cost and latency, re-check prompts, then switch before retirement

Anthropic's migration advice, with time for the change board.

D Switch production to the newest model today, and fix problems as they appear

Customers find the regressions instead of the tests.

THE SIGNAL

A retirement date plus a slow change process: start migration now, and test it properly.

> Corvane is deciding whether Tallyhawk's engineering assistant should move up from the Sonnet tier to the Opus tier.
>
> Five observations are on the table.

Which two observations most justify moving up a tier? Choose two.

A. On the evaluation set, Sonnet misses the bar on complex changes even at high effort

B. A newer Opus version was released last week

C. Most requests are short, simple classification tasks

D. The work is long-running and agentic, and each completed change is worth a great deal

E. The team wants to future-proof the system

◆ PAUSE AND DECIDE

Which two observations most justify moving up a tier? Choose two.

A On the evaluation set, Sonnet misses the bar on complex changes even at high effort

Measured evidence, after the cheaper dial was tried.

B A newer Opus version was released last week

Newness is not a requirement.

C Most requests are short, simple classification tasks

Points down a tier, not up.

D The work is long-running and agentic, and each completed change is worth a great deal

The workload Opus is built for, with value to justify the cost.

E The team wants to future-proof the system

A wish, not evidence.

THE SIGNAL

Step up on evidence and fit: measured shortfall after effort, and high-value agentic work.

In five lines

- **Choose the lightest model that clears the bar**, on capability, latency, cost and rate limits.
- **Requirements, shortlist, test, choose, route, re-test**: in that order.
- **Turn the effort dial** before jumping a tier; lower it where quality holds.
- **Mixed difficulty at volume: route.** Uniform difficulty: one model.
- **Every model change is a release:** regression set, comparison, rollback.

COMING UP

Next: system prompts, templates and prompting techniques.